NUTAN KUMAR NAIK

AI/ML Engineer · AI Observability & LLM Systems · Geoscientist

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PROFESSIONAL PROFILE

AI/ML engineer and geoscientist who builds production software. Leads research on Traccia, an open-source OpenTelemetry SDK for observing and governing AI agents, and is first author of two arXiv papers. Works in Python and TypeScript, from LLM evaluation engines to FastAPI services. M.Sc Geology (IIT Kharagpur), doctoral research at NIT Rourkela and training at ONGC and CMPDI, applied to subsurface, water and emissions problems in energy.

PROFESSIONAL EXPERIENCE

Lead AI Researcher, Algen AI Private Limited, Bengaluru 2025 – Present

- Lead research on Traccia, an OpenTelemetry-native observability, governance and compliance platform for AI agents, shipped as open-source Python and TypeScript SDKs (traccia-py, 130+ GitHub stars).
- Co-designed its telemetry pipeline, semantic guardrail assessment and hashed trace ledger, which turns runtime traces into tamper-resistant compliance evidence for regulations such as the EU AI Act.
- Built AgentStripes, an evidence-graded maturity model and assessment engine for AI agents (40 capabilities, LLM judges, deterministic scoring), and piloted it on 30 open-source agent repositories.

Doctoral Researcher, National Institute of Technology Rourkela 2021 – Present

- Estimate estuarine water quality from Sentinel-2 satellite imagery with machine learning, calibrated on field measurements (Bhitarkanika, Odisha).
- Industry projects: mine-water chemistry for MCL (15 coal mines) and IMFA; slope stability for SAIL iron-ore mines; resistivity surveys for Indian Railways; rock mass rating of an underground coal mine.

SELECTED PROJECTS · GITHUB.COM/NAIKNUTAN97

- **AgentStripes engine** (Python, LLM judges) Scores AI agents on 40 capabilities; fully reproducible.
- **Traccia reference agent** (Python, OpenTelemetry) Traced loan-approval agent; featured by SciPapermill.
- **AI Water Tracker** (FastAPI, Streamlit, LangChain) LLM agent app with REST API, traced with Traccia.

PUBLICATIONS

- *Traccia: An OpenTelemetry-Based Governance Platform for AI Systems.* arXiv:2607.14309, 2026. First author.
- *AgentStripes: An Evidence-Graded Maturity Model for AI Agents.* arXiv:2610.18603, 2026. First author.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL

AI/ML: PyTorch, scikit-learn, XGBoost, SHAP, LangChain, LLM APIs, LLM-as-judge evaluation

Engineering: OpenTelemetry, FastAPI, Streamlit, Next.js, Docker, GCP, GitHub Actions, Git, uv, SQLite

Geoscience & GIS: Well-log interpretation, hydrogeochemistry, MINEX, Sentinel-2, ESA SNAP, ArcGIS, QGIS

INDUSTRIAL TRAINING

- **ONGC, Kolkata.** Well-log interpretation, lithology identification and multi-well horizon correlation.
- **CMPDI, Bhubaneswar.** Borehole geophysical logging, drilling and 3D geological modelling in MINEX.

EDUCATION

Ph.D, Hydrogeochemistry & Remote Sensing, NIT Rourkela 2021 – Present

M.Sc, Geology, IIT Kharagpur · CGPA 8.54/10 2020

B.Sc, Government Degree College, Sundargarh · 72.67% 2018

ACHIEVEMENTS

IIT JAM 2018 AIR 108 · GATE 2020 AIR 234 · CSIR NET (LS) 2023 AIR 40 · Secretary, Technical Society, NIT Rourkela